University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2020, Part 2

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Suppose R is a commutative ring with identity and F is a free R-module of rank n. Show that for any R-module M the R-module $\mathrm{Hom}_R(F, M)$ is isomorphic to the direct sum of n copies of M.

2. Which of the following polynomials are irreducible? Explain, and factor the ones that are not irreducible.

(a) $X^3 + X + 1$ in $\mathbb{Q}[X]$;

(b) $X^4 + X^2 + 1$ in $\mathbb{Q}[X]$;

(c) $Y^2 - X^3 + X^2$ in $\mathbb{C}[X, Y]$.

3. Let R be an integral domain.

(a) Define what it means for an element of R to be irreducible.

(b) Define what it means for an element of R to be prime.

(c) Show that a prime element is irreducible.

4. Suppose F is a field, V is an F-vector space and $f : V \to V$ is a linear map.

(a) Show that if V has finite dimension then f is injective if and only if it is surjective.

(b) Give examples with V of infinite dimension, in which f is surjective but not injective.

(c) Give an example in which f is injective but not surjective. Justify both examples; you may use any results on dimension if they are stated clearly.

5. A minimal prime ideal of a ring R is a prime ideal that does not properly contain any other prime ideal. If F is a field, find all minimal prime ideals in the ring

$$R = F[x_1, x_2, x_3, x_4, x_5, x_6]/(x_1x_2, x_3x_4, x_5x_6).$$

6. Suppose K is a field and $a \in K$ is such that $X^3 - a$ and $X^2 + X + 1$ are irreducible in $K[X]$. If $L = K[X]/(X^3 - a)$, show that L contains exactly one root of the polynomial $X^3 - a$.

7. This problem has two parts.

(a) Find representatives in rational canonical form for all conjugacy classes in $GL_6(\mathbb{Q})$ with characteristic polynomial $(X^3 - 1)^2$.

(b) For each representative, give the Jordan normal form in $GL_6(\mathbb{C})$.